



DOMESTIC SERVICE ROBOT SENSING AND INSPECTION REQUIREMENTS

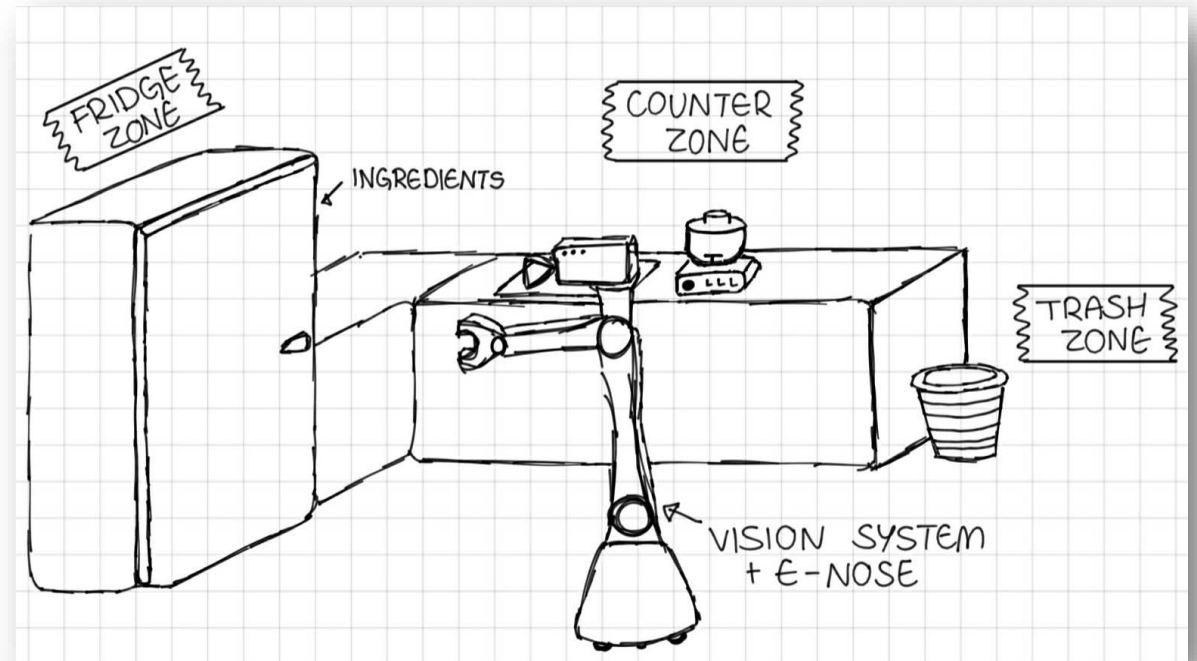
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OVERVIEW

Objective: The robot must prepare a smoothie, but the state of some ingredients is unknown. The robot must inspect ingredients before use.

Main requirements:

- ◆ Uncertainty - Freshness status is initially unknown for the robot.
- ◆ Sensing - Inspection actions (Vision + eNose) updates robot's knowledge.
- ◆ Dynamics - The food spoils over time (PDDL+)



PDDL DESIGN CHOICES

Uncertainty was modelled in classical PDDL by distinguishing reality from the robot's knowledge state by given predicates:

- ◆ is-fresh (?i) – physical state of an ingredient
- ◆ inspected (?I) – knowledge flag

Every sensing action has a high cost of 3 or 4 cost points forcing the robot to minimize sensing.

PDDL+ DESIGN CHOICES

Dynamics in PDDL+ was modelled by using three numeric fluents:

- ◆ (spoilage-level ?i - ingredient) - tracks the continuous degradation of the ingredient;
- ◆ (action-timer ?r - robot) - measures the elapsed time of a current task;
- ◆ (total-cost) - metric that helps planner to find well-optimized plan.

To ensure correct physical modelling, two main processes and events were used:

- ◆ spoilage-fridge - slowly increases food spoilage in the fridge (process)
- ◆ spoilage-outside - increases food spoilage outside the fridge (process)
- ◆ food-spoils - changes food state to unfresh if spoilage value exceeds the predefined limit.

REFLECTION

Main challenges:

- ◆ Modelling PDDL+ time to fit into ENHSP planner;
- ◆ Handling complexity of calculation in continuous time;
- ◆ Forcing the robot to execute tasks in shorter time.

What would I do differently:

- ◆ Easier, not overly complicated domain
- ◆ Less complex differential equations

Limitations:

- ◆ God-like Knowledge Paradox
- ◆ Precision vs Efficiency
- ◆ Goal-Oriented Constrains Limitation